



Course Name

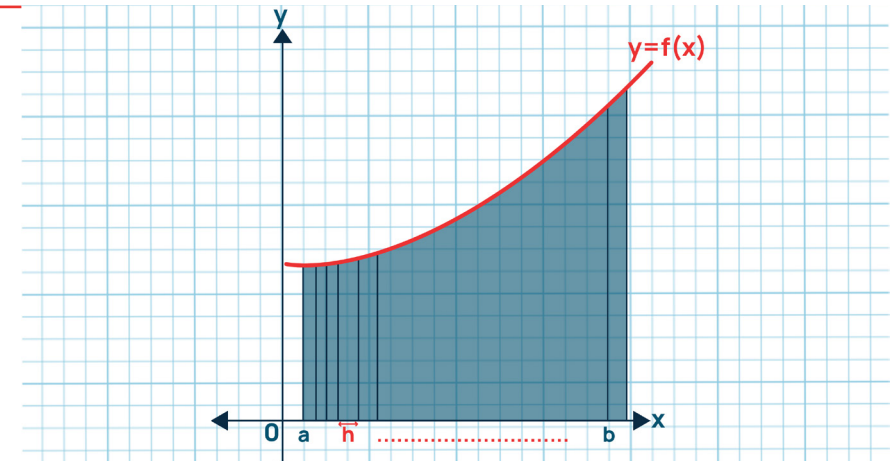
ELEMENTARY MATHEMATICS II (CALCULUS)

Topic: Integration

WEEK 7

IN SUMMARY

This week, you learned about the fundamental concept of integration in calculus. Integration is a mathematical technique that allows us to find the accumulation of quantities. It involves calculating the area under a curve or determining the change in a quantity. There are two main types of integrals: definite and indefinite. Definite integrals are used to find the exact value of the accumulated quantity between specific points on a graph, while indefinite integrals help us find a general formula for the accumulated quantity without specifying limits. The integration process results in an integral, and understanding these integrals is crucial for solving various mathematical problems.



BULLET POINT SUMMARY

- Understanding of the fundamental concept of integration in calculus, which involves calculating the accumulation of quantities, finding areas under curves, and solving differential equations.
- Differentiating between the two main types of integrals: definite, used to calculate the exact accumulated quantity between specific points, and indefinite, which provides a general formula for accumulated quantities.
- Learning specific integration techniques, including the Power Rule for functions of the form ' x^n ,' trigonometric integrals, and exponential integrals.
- Exploring advanced integration methods, such as integration by sub

CASE STUDY

Title: Application of Integration in Structural Engineering:

Introduction: In this case study, we delve into the application of integration in the field of structural engineering. Integration plays a pivotal role in analysing and optimising structural systems. Understanding its practical use is crucial for designing safe and efficient structures.

Objective: The objective of this case study is to investigate how integration techniques, including both definite and indefinite integrals, are employed in structural engineering to assess the behaviour of structural components and systems. By examining real-world structural engineering

Table of Integration

$\int k \, dk = Kx + c$	$\int \sec^2 x \, dx = \tan x + c$
$\int x^n \, dx = \frac{1}{n+1} x^{n+1} + I_{+C, n \neq -1}$	$\int \sec x \tan x \, dx = \sec x + c$
$\int x^{-1} \, dx = \int \frac{1}{x} \, dx = \ln x + C$	$\int \csc x \cot x \, dx = -\csc x + c$
$\int \frac{1}{ax+b} \, dx = \frac{1}{a} \ln ax+b + c$	$\int \csc^2 x \, dx = -\cot x + c$
$\int \ln(x) \, d^2c = x \ln(x) - x + c$	$\int \tan x \, dx = \ln \sec x + c$
$\int e^x \, dx = e^x + c$	$\int \sec x \, dx = \ln \sec x + \tan x + c$
$\int \cos x \, dx = \sin x + C$	$\int \frac{1}{a^2 + u^2} \, dx = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + c$
$\int \sin x \, dx = -\cos x + c$	$\int \frac{1}{\sqrt{a^2 - u^2}} \, dx = \sin^{-1}\left(\frac{u}{a}\right) + c$

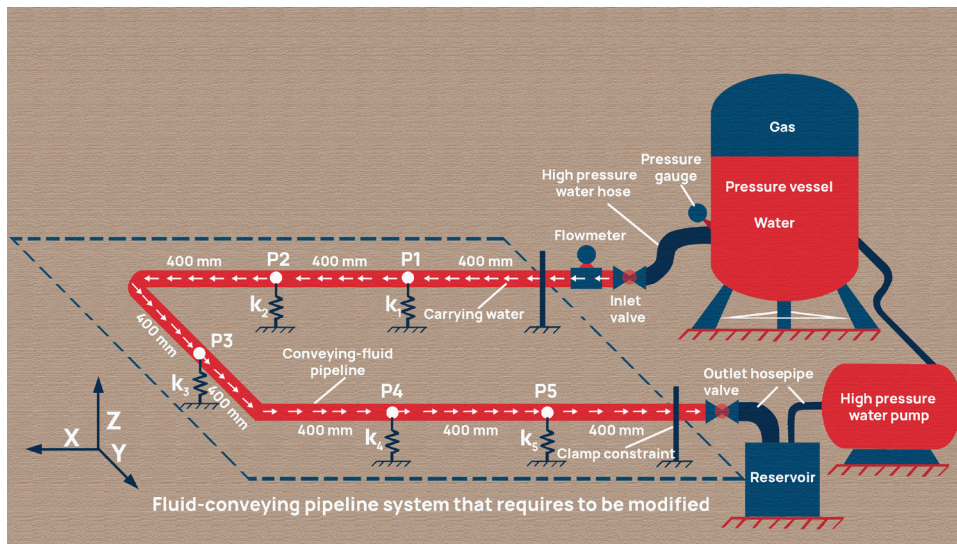
CASE STUDY CONT'D

projects, we aim to demonstrate the significance of integration in solving engineering problems.

Methodology: We will select a specific structural engineering project where integration techniques are essential. This case study may involve the analysis of a bridge's load-bearing capacity or the determination of stresses and strains in a building's framework. Data and information related to the project, as well as the integration techniques applied, will be collected. A comprehensive analysis will be conducted to assess how integration was used to optimise the design and analyse the structural behaviour.

QUESTIONS TO PONDER

- How can integration techniques be further optimised to enhance the efficiency and accuracy of structural analysis in engineering design?
- In what ways can engineering education better emphasise the practical applications of integration to prepare future structural engineers for real-world problem-solving challenges?



SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

- Proficiency in solving problems involving integration.
- Differentiating between indefinite and definite integrals
- Application of integration techniques in mathematical problem-solving.
- Critical thinking and problem-solving skills.
- An understanding of integration's significance in engineering applications.
- Analytical thinking and problem-solving in complex systems.

PERSONAL REFLECTION

Studying integration this week deepened my understanding of its fundamental role in mathematics and its wide-ranging applications in engineering. I gained valuable problem-solving skills and recognised how mathematical concepts directly translate to practical engineering solutions.